Trigonometry

Section 1: Exact Trig Ratios & Reference Angles

1 Find the exact value of $\sin(210^\circ)$.

a. $-\frac{1}{2}$

b. $\frac{1}{2}$

c. $-\frac{\sqrt{3}}{2}$

d. $\frac{\sqrt{3}}{2}$

2 Find the exact value of $\cos\left(\frac{3\pi}{4}\right)$.

a. $-\frac{\sqrt{2}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. $-\frac{1}{2}$

d. $\frac{\sqrt{3}}{2}$

3 If θ is reflex and $\tan(\theta) = \frac{4}{3}$, find the exact value of $\sin(\theta)$.

a. $-\frac{4}{5}$

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

Section 2: Fundamental Trigonometric Identities

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4 Simplify the expression $(1-\cos^2\theta)(1+\cot^2\theta)$ for all valid θ.

a. 1

b. $\sin^2\theta$

c. $\tan^2\theta$

d. $\sec^2\theta$

5 Prove the identity $\dfrac{\sin\theta}{1-\cos\theta}+\dfrac{1-\cos\theta}{\sin\theta}\equiv 2\csc\theta$. What is the common numerator after combining fractions?

a. $2(1-\cos\theta)$

b. $\sin^2\theta+\cos^2\theta$

c. $2\sin\theta$

d. $1-2\cos\theta$

6 Given $\cos\theta=-\frac{5}{13}$ and $90^\circ<\theta<180^\circ$, find $\tan\theta$.

a. $-\frac{12}{5}$

b. $\frac{12}{5}$

c. $-\frac{5}{12}$

d. $\frac{13}{12}$

Section 3: Solving Trigonometric Equations

7 Solve $2\sin(2x)=1$ for $0^\circ\le x\le 180^\circ$.

a. $x=15^\circ$ and $x=75^\circ$

b. $x=30^\circ$ and $x=150^\circ$

c. $x=15^\circ$ and $x=105^\circ$

d. $x=45^\circ$ and $x=135^\circ$

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8 Solve $2\cos^2\theta - 5\cos\theta + 2 = 0$ for $0^\circ \le \theta \le 360^\circ$.

a. $\theta = 60^\circ$ and $\theta = 300^\circ$

b. $\theta = 30^\circ$ and $\theta = 330^\circ$

c. $\theta = 60^\circ$ and $\theta = 120^\circ$

d. $\theta = 45^\circ$ and $\theta = 315^\circ$

9 Solve $3\tan(x) = 2\sin(x)$ for $0 < x < 2\pi$.

a. $x = \pi$ (excluded) or $\cos x = 3/2$ (no solution)

b. No solutions in $(0, 2\pi)$

c. $x = \pi/3$ and $5\pi/3$

d. $x = \pi/6$ and $5\pi/6$

Section 4: Trigonometric Graphs & Transformations

10 State the amplitude and period of the function $y = 3\sin(2x) - 1$.

a. Amplitude $= 3$, Period $= \pi$

b. Amplitude $= 3$, Period $= 2\pi$

c. Amplitude $= 2$, Period $= \pi$

d. Amplitude $= 4$, Period $= \pi$

11 Find the range of the function $f(x) = 4 - 3\cos(x)$ for all real x.

a. $[1, 7]$

b. $[-1, 7]$

c. $[1, 4]$

d. $[-3, 3]$

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12 Find the number of solutions to the equation $\sin(2x) = \cos(x)$ in the interval $0 \leq x \leq 2\pi$.

a. 4 solutions

b. 2 solutions

c. 3 solutions

d. 6 solutions

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